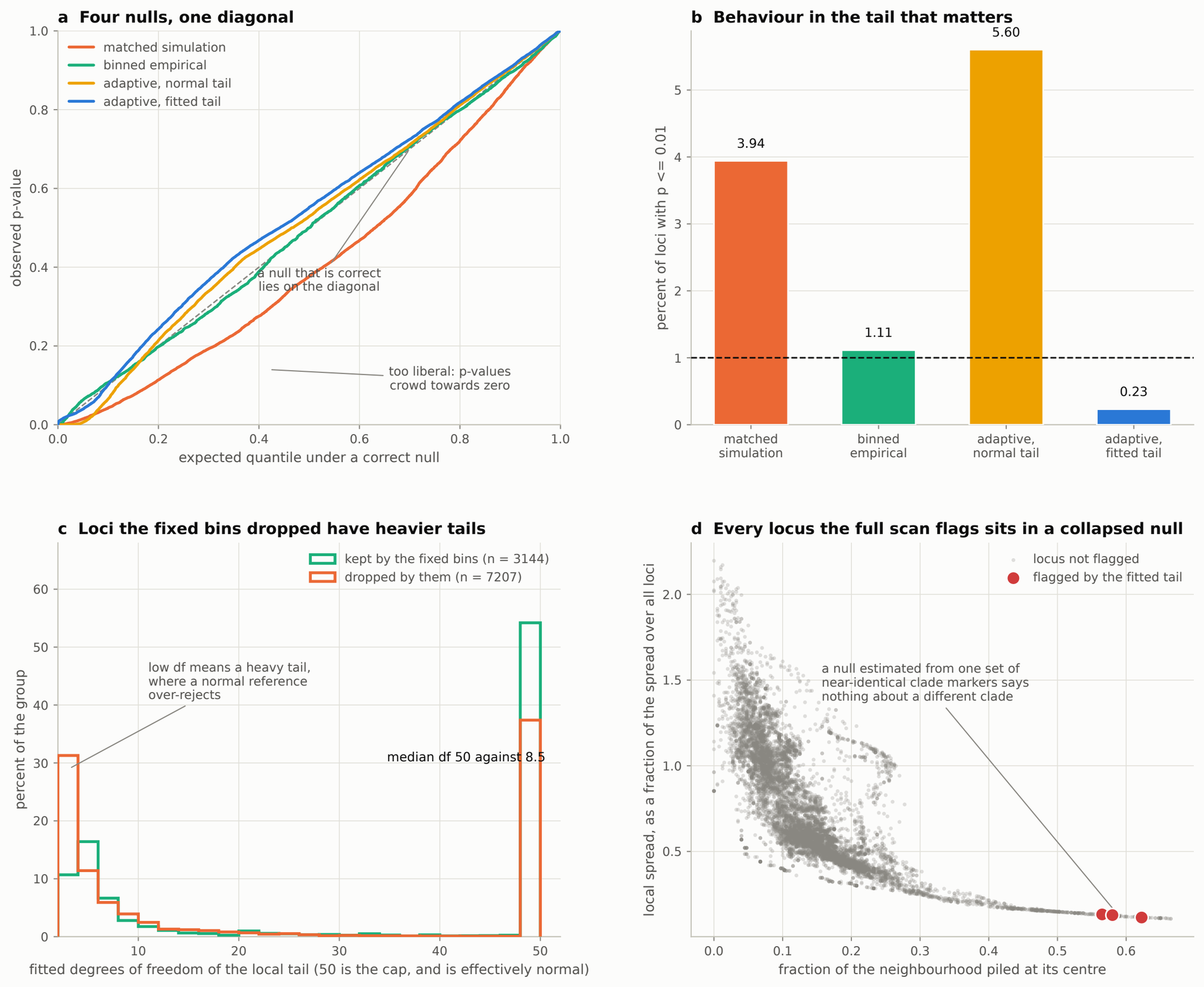

a Four nulls, one diagonal
matched simulation
binned empirical
adaptive, normal tail
adaptive, fitted tail
observed p-value
expected quantile under a correct null
a null that is correct
lies on the diagonal
too liberal: p-values
crowd towards zero
b Behaviour in the tail that matters
percent of loci with p <= 0.01
3.94
1.11
5.60
0.23
matched simulation
binned empirical
adaptive, normal tail
adaptive, fitted tail
c Loci the fixed bins dropped have heavier tails
kept by the fixed bins (n = 3144)
dropped by them (n = 7207)
percent of the group
low df means a heavy tail,
where a normal reference
over-rejects
median df 50 against 8.5
fitted degrees of freedom of the local tail (50 is the cap, and is effectively normal)
d Every locus the full scan flags sits in a collapsed null
locus not flagged
flagged by the fitted tail
local spread, as a fraction of the spread over all loci
a null estimated from one set of
near-identical clade markers says
nothing about a different clade
fraction of the neighbourhood piled at its centre